PI-DeepONet for Homogeneous system trajectory prediction

$$\begin{cases} \dfrac{dx}{dt} = F(x, \mu) \\ x(0) = x_0 \end{cases} \longrightarrow \text{DeepONet predicts: } \phi(\mu, x_0)(t)$$

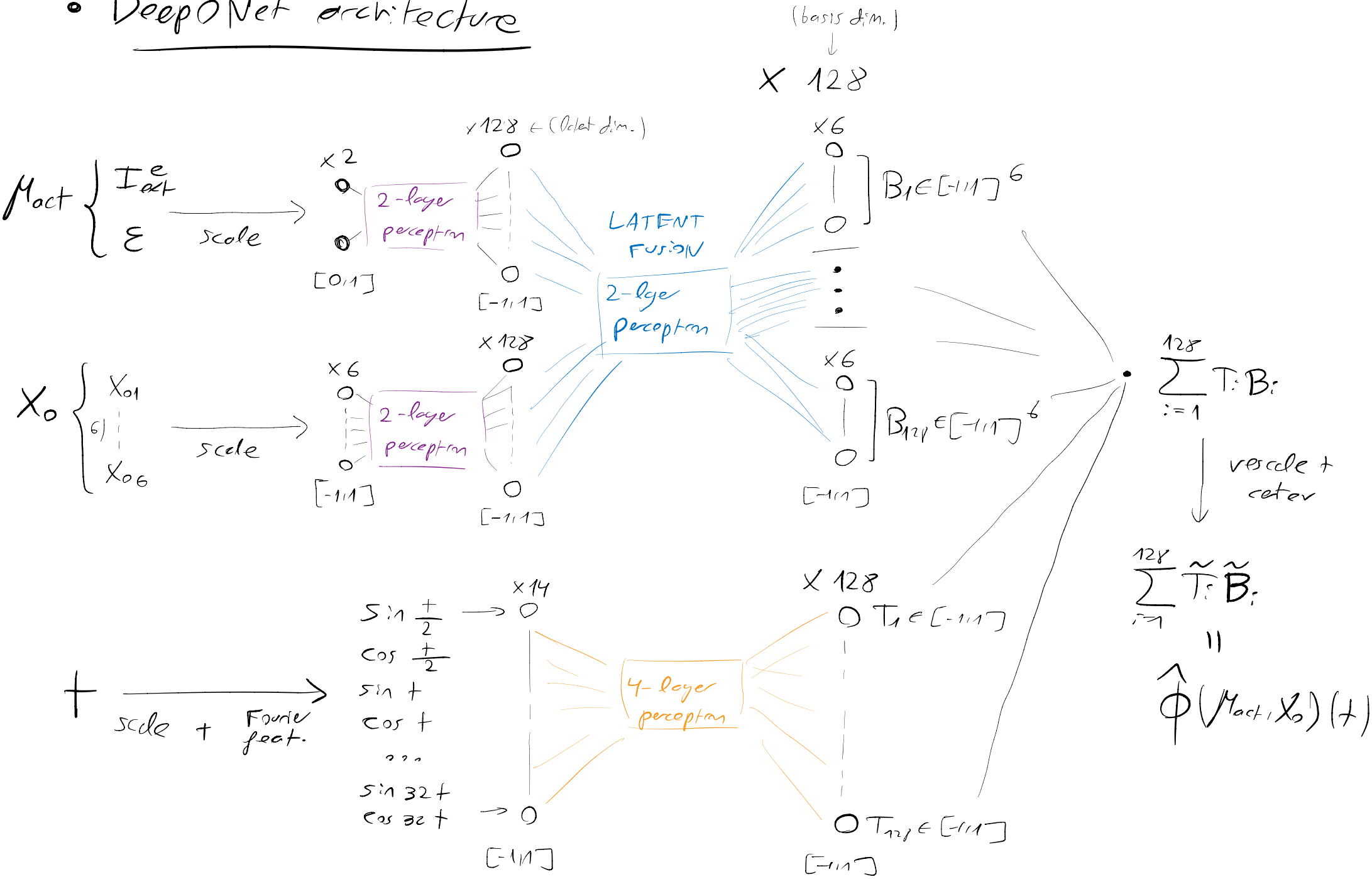
$x \in \mathbb{R}^6$, $\mu \in \mathbb{R}^{15}$

Homogeneous system experiment:

- $W = [[1]]$, $N_{pop} = 1$ (fixed)
- $\mu = [\ldots, F_{ext}^e, \varepsilon]$ ← All fixed except $F_{ext}^e \in [0, 16]$, $\varepsilon \in [0, 35]$
- $x_0 \in N(0 \in \mathbb{R}^6, \sigma^2)$, σ^2 small
- $t \in [0, 10]$

$\bigcirc \equiv$ tanh perceptron

• DeepONet architecture



- <u>Loss backpropagation</u>

supervision

1) supervision loss: $L_{data} = \left\| \hat{\phi}(\mu_{act}, x_0)(t) - \phi_{integrated}(\mu_{act}, x_0)(t) \right\|^2$

2) IC loss: $L_{ic} = \left\| \hat{\phi}(\mu_{act}, x_0)(0) - x_0 \right\|^2$

3) ODE loss: $L_{ode} = \left\| \dfrac{d\hat{\phi}/dt - F(\hat{\phi})}{\text{scales}} \right\|^2$

variables have very different scales!

captures physical trajectories

4) variational eq. loss: $L_{var} = \left\| \dfrac{d(D\hat{\phi})/dt - DF \cdot D\hat{\phi}}{\text{scales}} \right\|^2$

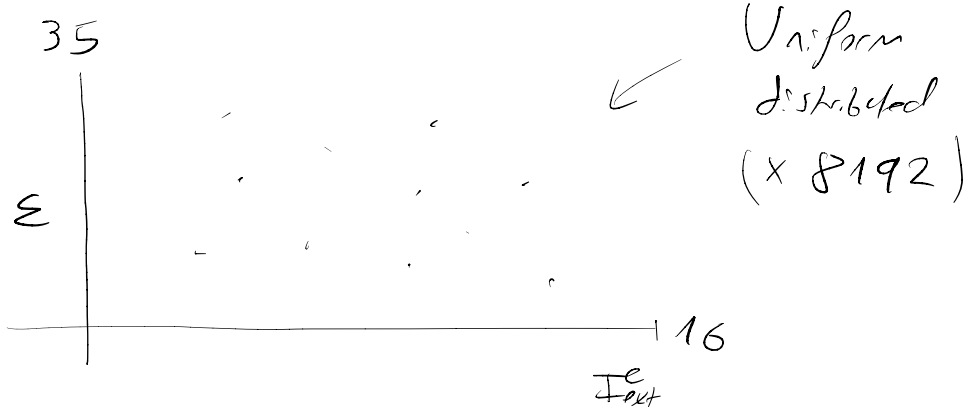
captures variability on physical trajectories

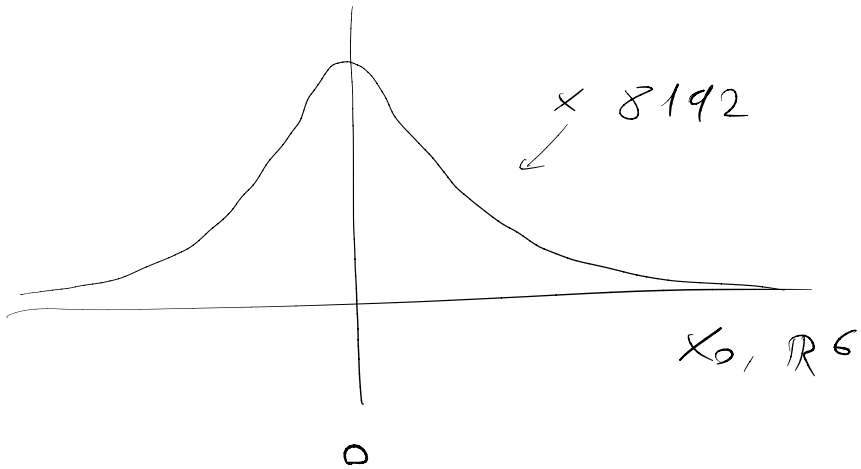
REQUIRED FOR LYAPUNOV

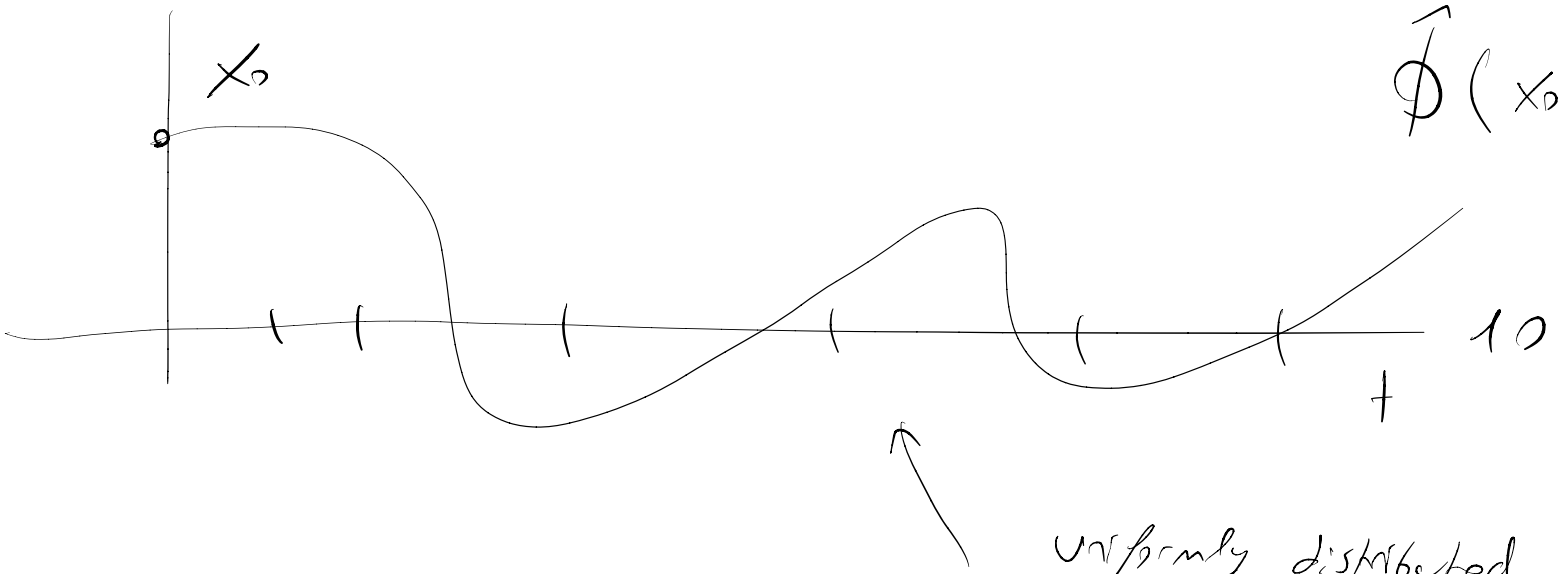
- Training

$$L = \underbrace{\underbrace{L_{data} + 10 \cdot L_{ic}}_{\text{Phase I of training: supervised pre-training}} + 0.1 \cdot L_{ODE} + 0.01 \cdot L_{var}}_{\text{phase II of training: PI fine-tuning}}$$

- Sampling:







$\hat{\phi}(x_0, \mu_{data})(t)$

$\times$ 8192

uniformly distributed ($\times$ 512)

(for each trajectory)

- optimizer

Adam (adaptative moment) $\begin{cases} \bullet \text{ Momentum } (1^{st} \text{ moment, mean } m) \\ \bullet \text{ Adaptative learning rate } (2^{nd} \text{ moment, variance } v) \end{cases}$

+ scheduled decreasing η + clipping

$$w_{t+1} = w_t - \frac{\eta}{\sqrt{\hat{v}_t} + \varepsilon} \hat{m}_t, \qquad \eta \to \frac{\eta}{2} \text{ every } N_{steps}$$

Phase I $\longrightarrow$ Phase II

Phase I

- 100 000 resamples
- 32-minibatch under supervision:

$$\mu_{\text{data}}, x_0, [t_{11}, \dots, t_{12}] \times 32$$

$$\downarrow$$

NN parameter update

- $\eta_0 = 10^{-3}$, $\eta \to \eta/2$ every 20 000 steps

Phase II

- 50 000 resamples
- 16-minibatch under supervision + PI

  16-minibatch only PI
- $\eta_0 = 10^{-5}$, $\eta \to \eta/2$ every 10 000 steps